CSE 270 Sports Analytics

Quiz 1

Name: _____

Problem 1

What is the correct way of installing a library in R?

```
A) install.packages("devtools")
B) install.packages(devtools)
C) library("devtools")
D) library(devtools)
```

Problem 2

Do these two codes perform the same action ?

```
a <- c(1:10)
a <- c(1,10)

A) TRUE
B) FALSE
```

Problem 3

```
class_schedules <- c("MWF 12:30-13:20","TT 9:00-10:15", "MWF 08:30-09:20", "MWF 10:30-11:20")
names(class_schedules) <- c("CSE270", "CS222", "CS112", "CS246")
```

a) Get the schedule for CSE 270,
Pick all the correct options.

```
A) class_schedules["CSE270"]
B) class_schedules[0]
C) class_schedules[1]
D) class_schedules$CSE_270
```

b) Get the schedule for all classes except ***CSE 270***,
Pick all the correct options.

```
A) class_schedules[-c("CSE270")]
B) class_schedules[2:4]
C) class_schedules[c(2,4)]
D) class_schedules[names(class_schedules) != "CSE270"]
```

Problem 4

Which of these code chunks will create a 3x3 matrix
and fill it with numbers from 1 to 9.
It should look like this.

```
##      [,1] [,2] [,3]
## [1,]    1    2    3
## [2,]    4    5    6
## [3,]    7    8    9
```

Function matrix() **Arguments**

data
an optional data vector (including a list or expression vector)

nrow
the desired number of rows.

ncol
the desired number of columns.

byrow
logical. If FALSE (the default) the matrix is filled by columns, otherwise the matrix is filled by rows.

Pick all that apply

```
A) one_to_nine <- matrix(1:9)
B) one_to_nine <- matrix(1:9, byrow=TRUE, nrow=3)
C) one_to_nine <- matrix(c(1,9), byrow=TRUE, nrow=3)
D) one_to_nine <- matrix(rbind(c(1:3), c(4:6), c(7:9)))
```

Problem 5

a) Write down the commands to change the column names and row names of
the matrix ***one_to_nine***, set the column names to ("Lazio", "Roma" , "Juventus") and
the row names to ("2022", "2021", "2020")

```
# Write down your code here, should not take more than two lines
#
#
```

b) The matrix displays the standings for the teams during the last three years, write a command to get Lazio's final position in 2020, can you get it in multiple ways ?

```
# Write down your code here, there are at least 3 ways to do that
#
#
#
```

c) Write down the output of this code chunk

```
rowSums(one_to_nine)
```

```
# The output should not be more than 2 lines
#
#
```

Problem 6

a) Write a code chunk to convert the matrix ***one_to_nine*** into a dataframe and name that dataframe ***standings_2020_2022***.

```
# Write doum your code here, should take no more than 1 line
#
```

b) Write a code chunk to add one more column to the data frame, name it "Napoli" and add values (4, 2, 3) for the values of that column.

```
# Write doum your code here, should take no more than 2 lines
#
#
```

c) Write a code chunk to add one more row to the data frame, name it 2019 and add the values, (3,4,2,5)

```
# Write doum your code here, should take no more than 1 line
#
```

Problem 7

a) R has a built in dataframe called ***iris*** that contains some information about Species.
Write down the commands to get the data frame and show the last 3 observations of the data frame.

```
# Write doum your code here, should take no more than 2 lines
#
#
```

b) Write a code chunk to create 2 new variables for the dataframe iris and call them ***Sepal.Area*** and ***Petal.Area*** and fill in the values with the area(width x length) of both variables correspondingly,

```
# Write doum your code here, should take no more than 2 lines
#
#
```

c) Write a code chunk to filter out the data frame ***iris*** to only include the species for which the variable ***Sepal.Area*** is greater than 10 and the variable ***Petal.Area*** is less than 1.

```
# Write doum your code here, should take no more than 1 line
#
```